

Ahmet Keles

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EDUCATION

The University of Texas at Dallas

Bachelor of Science, Computer Science

• **Coursework:** Data Structures & Algorithms, Computer Architecture, UNIX, Java OOP & Data Structures

Dec 2027

Richardson, TX

SKILLS

Languages: Java, C++, Python, C#, JavaScript, SQL, Assembly

Frameworks/Web: Spring Boot, React.js, ASP.NET/MVC, HTML5, CSS, Bootstrap, jQuery

Data & Messaging: PostgreSQL, MSSQL, Redis, Kafka, RabbitMQ

Tools/Infrastructure: Git, Docker, Linux, REST APIs, gRPC, CI/CD

Cloud: AWS, Azure

PROFESSIONAL EXPERIENCE

Keles Group Inc.

Jun 2026 - Aug 2026

Superintendent Intern

• Supported superintendent teams across Wylie and Grand Prairie on construction projects totaling over \$10.2M by assisting with field operations and project execution, helping support day-to-day progress across multiple active sites.

DTF Order

Jun 2024 - Aug 2024

Customer Experience Intern

Richardson, TX

• Worked in a fast-paced customer experience environment at DTF Order, supporting customer communication and day-to-day service needs to help maintain responsive order support.

PROJECTS

Event-Driven Commerce Platform [GitHub: event-driven-commerce](#)

2026

- To coordinate orders, inventory, and payments without tightly coupling services, built a 3-service Kafka-based commerce backend with isolated PostgreSQL databases, enabling independent scaling and failure recovery.
- To make at-least-once delivery safe under retries and concurrency, implemented the Transactional Outbox pattern, idempotent consumers, saga compensation, optimistic locking, bounded retries/DLTs, and FOR UPDATE SKIP LOCKED, reducing duplicate-processing and race-condition risk.
- To verify behavior under production-like conditions, developed 345 unit, integration, and end-to-end tests with Testcontainers across Kafka and three PostgreSQL instances, while Docker Compose and Flyway kept local environments reproducible and schemas consistent.

NovaSearch - AI-Powered Distributed Search Engine [GitHub: novasearch](#)

2026

- To make search understand both exact terms and user intent, designed NovaSearch around hybrid keyword and semantic retrieval with RAG, creating a clear path from raw documents to grounded AI-generated answers.
- To support grounded LLM responses, planned a LangChain RAG pipeline for document chunking, embeddings, pgvector indexing, reranking, and source-cited REST API responses, targeting stronger retrieval relevance and traceability.
- To keep the AI system measurable and production-oriented, defined Python/FastAPI, Redis, Docker, Git, and evaluation workflows around retrieval quality, P95 latency, cache hit rate, and inference cost.

Distributed Job Scheduler & Workflow Engine [GitHub: distributed-job-scheduler](#)

2026

- To keep background work reliable across worker failures, designed a distributed scheduler that prioritized and coordinated jobs across multiple workers, with architecture centered on fault-tolerant execution.
- To prevent lost or duplicated work, planned worker heartbeats, leases, distributed locking, retries, timeouts, dead-letter recovery, persistent execution history, and cancellation flows.
- To validate scale and operability, specified REST/WebSocket monitoring, load tests, Docker-based workers, and AWS deployment scenarios for worker scaling and failure recovery.

Distributed Object Storage System [GitHub: distributed-object-storage](#)

2026

- To keep large files available despite node failures, designed a distributed object storage service that distributed data across multiple nodes with replication and recovery-oriented architecture.
- To improve placement, resiliency, and lookup efficiency, planned consistent-hash chunk placement, configurable replication, node health checks, background repair, Redis-assisted metadata caching, and gRPC communication.
- To verify durability and performance, defined node-loss and corruption tests plus measurements for upload/download throughput, recovery time, storage overhead, and metadata latency.

EXTRACURRICULAR

Association for Computing Machinery (ACM)

Sep 2023 - Present

Member

• Expanded practical software-engineering exposure by participating in ACM meetings, technical workshops, and collaborative programming activities beyond coursework.

Sachse Public Library

2022

Volunteer

• Supported daily library operations by organizing materials, assisting an average of three patrons per day, and helping with public events and community outreach.

Raindrop Foundation

2021 - 2024

Volunteer

• Supported Raindrop Foundation community initiatives as a volunteer from 2021 to 2024, contributing consistent service over a three-year period.